

Week 9 — Lecture notes

System analysis with the Laplace transform

The web version of these notes is at tchaffey.com/elec2302.

Lecture outline

Last week:

- Laplace transform converges for unstable poles.
- Unilateral Laplace transform integrates from 0 to ∞ , and allows us to take care of initial conditions.

This week:

- Laplace transform & stability
- Inverse Laplace transforms using partial fractions
- System zeros.

Laplace transform & stability

Previously, we have used the Fourier transform of the impulse response as a transfer function. The transfer function lets us compute the output by multiplication with the input:

Time domain

$$y(t) = \underbrace{h(t)}_{\text{impulse response}} * \underbrace{u(t)}_{\text{input}}$$

Frequency domain

$$Y(j\omega) = \underbrace{H(j\omega)}_{\text{transfer function}} \underbrace{U(j\omega)}_{\text{input}}$$

This only works when the Fourier transform integral converges, but we saw last week for the Laplace transform:

$$y(t) = h(t) * u(t) \rightsquigarrow Y(s) = H(s) U(s).$$

So we can reuse our definition of the transfer function:

The transfer function of a system is the Laplace transform of its impulse response.

Our previous definition of a transfer function, the Fourier transform of the impulse response, is usually known as the frequency response.

The Laplace version lets us calculate the transfer function of an unstable system, whose impulse response grows exponentially:

$$\mathcal{L}(e^{pt})(s) = \frac{1}{s-p}, \quad p > 0.$$

An unstable impulse response gives a right half plane pole.

We can now complete our stability catalogue:

Exponentially stable:	poles in the open left half plane
Marginally stable:	poles in the closed left half plane, at least one pole on the imaginary axis, no repeated poles on the imaginary axis.
Unstable:	at least one pole in the right half plane.
BIBO stable:	$\int_{-\infty}^{\infty} h(t) dt < \infty$
Exponential stability $\Rightarrow$ BIBO stability.	
BIBO stable & no pole/zero cancellations $\Rightarrow$ exponential stability.	

We can check pole locations by factorising the denominator of the transfer function:

$$\frac{s-1}{\underbrace{(s+1)}_{\text{stable}} \underbrace{(s^2+s+1)}_{\text{stable}} \underbrace{(s-2)}_{\text{unstable}}}$$

To calculate the corresponding time domain response, however, we need to separate into components in the transform table. This is called Partial Fraction Expansion.

Inverse Laplace transforms using partial fractions

Example 1: simple poles

$$\frac{1}{s^2+3s+2} = \frac{1}{(s+1)(s+2)}.$$

The residue or cover-up method splits this into two terms:

$$\frac{A}{s+1} + \frac{B}{s+2}.$$

To find A : cover up the $s+1$ part:

$$\frac{1}{\boxed{s+1} (s+2)}$$

and plug in the value of the covered pole: $s = -1$.

This gives $A = \frac{1}{1} = 1$. (This is called a residue in complex analysis).

To find B : repeat with the $s+2$ pole:

$$\frac{1}{(s+1) \boxed{s+2}}$$

Plug in $s = -2$ to get $B = \frac{1}{-1} = -1$.

So we have

$$\frac{1}{s+1} - \frac{1}{s+2}.$$

Let's check this is correct:

$$\frac{(s+2) - (s+1)}{(s+1)(s+2)} = \frac{1}{(s+1)(s+2)},$$

which is what we started with.

We can now calculate the inverse Laplace transform by matching with table entries: in this case,

$$\frac{1}{s+p} \rightsquigarrow e^{-pt}.$$

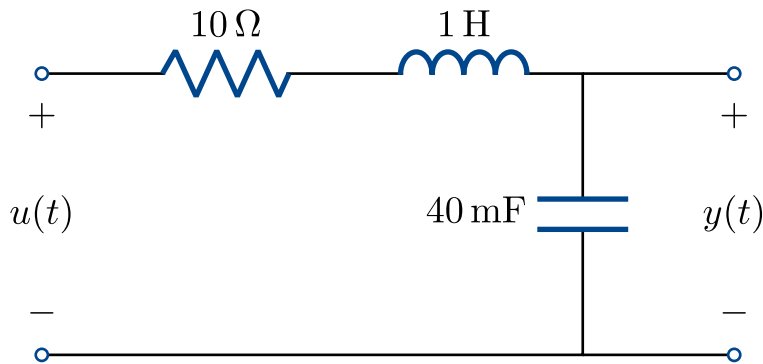
So the time domain signal corresponding to

$$\frac{1}{s^2 + 3s + 2} = Y(s)$$

is

$$y(t) = e^{-t} - e^{-2t}, \quad t \geq 0.$$

Example 2: repeated pole



Transfer function is

$$G(s) = \frac{25}{s^2 + 10s + 25}.$$

Input is a step, so $U(s) = \frac{1}{s}$.

$$\begin{aligned} Y(s) &= G(s)U(s) \\ &= \frac{25}{s(s+5)^2} = \frac{A}{s} + \frac{B}{s+5} + \frac{C}{(s+5)^2}. \end{aligned} \quad (1)$$

We can find A & C using cover-up:

$$A = \left. \frac{25}{(s+5)^2} \right|_{s=0} = 1$$

$$C = \left. \frac{25}{s} \right|_{s=-5} = -5.$$

For B , from (1) we have

$$25 = A(s+5)^2 + Bs(s+5) + Cs.$$

The s^2 coefficient is $A + B = 0$, so $B = -1$.

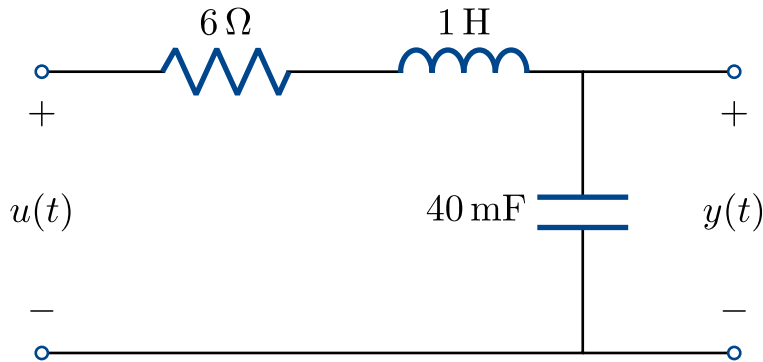
So we have

$$Y(s) = \frac{1}{s} - \frac{1}{s+5} - \frac{5}{(s+5)^2}.$$

Using the transform table,

$$y(t) = 1 - e^{-5t} - 5te^{-5t}, \quad t \geq 0.$$

Example 3: complex pair



Transfer function is

$$G(s) = \frac{25}{s^2 + 6s + 25}.$$

Input is a step, so $U(s) = \frac{1}{s}$.

$$Y(s) = \frac{25}{s \underbrace{(s^2 + 6s + 25)}}.$$

$$b^2 - 4ac = 36 - 100 < 0.$$

Underdamped, so the expansion is

$$Y(s) = \frac{A}{s} + \frac{Bs + C}{s^2 + 6s + 25}.$$

Use cover-up to find A :

$$A = \left. \frac{25}{s^2 + 6s + 25} \right|_{s=0} = 1.$$

Coefficient matching:

$$25 = (s^2 + 6s + 25) + s(Bs + C)$$

$$\begin{aligned}s^2 \text{ coefficient: } 0 &= 1 + B. & B &= -1 \\s \text{ coefficient: } 0 &= 6 + C. & C &= -6.\end{aligned}$$

$$Y(s) = \frac{1}{s} - \frac{(s+6)}{s^2 + 6s + 25}.$$

Now complete the square:

$$\begin{aligned}Y(s) &= \frac{1}{s} - \frac{s+6}{(s+3)^2 + 16} \\&= \frac{1}{s} - \frac{s+3}{(s+3)^2 + 16} - \frac{3}{(s+3)^2 + 16} \\&= \frac{1}{s} - \frac{s+3}{(s+3)^2 + 16} - \frac{3}{4} \frac{4}{(s+3)^2 + 16}.\end{aligned}$$

Now match with the table:

$$y(t) = (1 - e^{-3t} (\cos 4t + 0.75 \sin 4t)), \quad t \geq 0$$

Example 4

$$Y(s) = \frac{2s}{(s+2)(s^2+4)} = \frac{A}{s+2} + \frac{Bs+C}{s^2+4}.$$

Cover-up gives

$$A = \left. \frac{2s}{(s^2+4)} \right|_{s=-2} = \frac{-4}{8} = -\frac{1}{2}.$$

Coefficient matching:

$$2s = A(s^2 + 4) + (Bs + C)(s + 2).$$

$$s^2 \text{ coefficient: } 0 = A + B. \quad B = \frac{1}{2}.$$

$$s^0 \text{ coefficient: } 0 = 4A + 2C. \quad C = 1.$$

$$Y(s) = \frac{-\frac{1}{2}}{s+2} + \frac{\frac{1}{2}s+1}{s^2+4}$$

$$y(t) = -\frac{1}{2}e^{-2t} + \frac{1}{2}\cos 2t + \frac{1}{2}\sin 2t, \quad t \geq 0.$$

Example 5

$$Y(s) = \frac{s^2}{s^2 + 6s + 25}$$

$$\begin{aligned}Y(s) &= \frac{s^2 + 6s + 25}{s^2 + 6s + 25} - \frac{(6s + 25)}{s^2 + 6s + 25} \\&= 1 - \frac{6(s+3) + 7}{(s+3)^2 + 16} \\&= 1 - 6 \frac{s+3}{(s+3)^2 + 16} - \frac{7}{4} \frac{4}{(s+3)^2 + 16}\end{aligned}$$

$$y(t) = \delta(t) - e^{-3t} \left(6 \cos 4t + \frac{7}{4} \sin 4t \right), \quad t \geq 0.$$

Summary table of decompositions

$$\frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}.$$

$$\frac{25}{s(s+5)^2} = \frac{A}{s} + \frac{B}{s+5} + \frac{C}{(s+5)^2}.$$

$$\frac{25}{s(s^2+6s+25)} = \frac{A}{s} + \frac{Bs+C}{s^2+6s+25}.$$

$$\frac{2s}{(s+2)(s^2+4)} = \frac{A}{s+2} + \frac{Bs+C}{s^2+4}.$$

$$\frac{s^2}{s^2+6s+25} = 1 + \frac{As+B}{s^2+6s+25}.$$

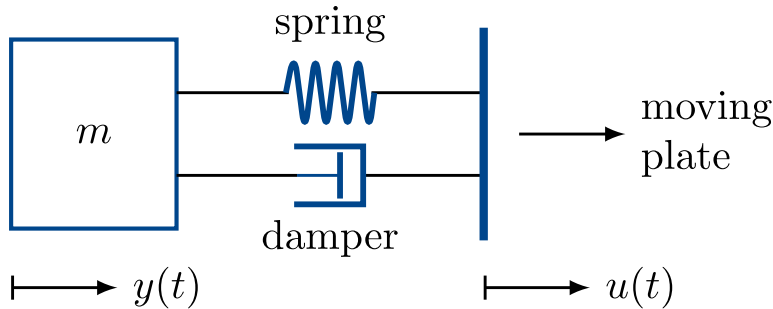
System zeros

So far, we have only looked at the poles of the transfer function (the roots of the denominator). In the time domain, these give decaying or growing exponentials.

What about the roots of the numerator? These are called the zeros of the system.

Intuitively, zeros tell us certain inputs are blocked by the system and don't appear at the output. Let's see how this works with a couple of examples. We'll use mechanical examples as they are a lot easier to visualise in this case, but a good exercise is to find equivalent RLC circuit examples.

Example 6



Damper force is proportional to velocity:

$$f_d = k_d(\dot{u} - \dot{y}).$$

Spring force is proportional to $y - u$:

$$f_s = k_s(u - y).$$

Newton's 2nd law:

$$\begin{aligned} \ddot{y}m &= f_d + f_s \\ &= k_d(\dot{u} - \dot{y}) + k_s(u - y). \end{aligned}$$

Question: can we move the plate so that the damper force and spring force cancel and the mass stays still?

Intuitively: if the spring is one unit away from rest, the spring pulls it back and the damper resists this motion. For the forces to match, the closer the plate, the slower it must move.

Let's try and find this input using the Laplace transform.

$$m\ddot{y} + k_d\dot{y} + k_s y = k_d\dot{u} + k_s u$$

$$\begin{aligned} m(s^2 Y(s) - \dot{y}(0) - sy(0)) + k_d(sY(s) - y(0)) + k_s Y(s) \\ = k_d(sU(s) - u(0)) + k_s U(s). \end{aligned}$$

Let's assume $\dot{y}(0) = y(0) = 0$. Then we have:

$$Y(s) = \frac{(k_d s + k_s)U(s) - k_d u(0)}{ms^2 + k_d s + k_s} \quad \left(\text{note the zero at } s = -\frac{k_s}{k_d} \right).$$

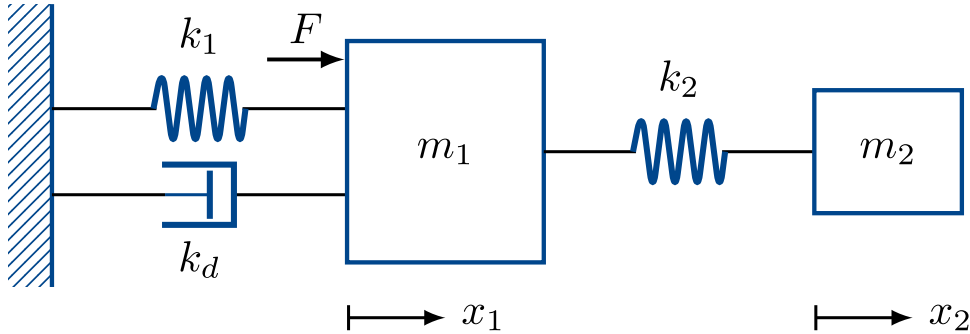
So if

$$U(s) = \frac{k_d u(0)}{k_d s + k_s} = \frac{u(0)}{s + \frac{k_s}{k_d}}, \quad Y(s) = 0.$$

In the time domain, this is

$$u(t) = u(0)e^{-\frac{k_s}{k_d}t}.$$

Example 7



$$m_1\ddot{x}_1 + k_d\dot{x}_1 + k_1x_1 + k_2(x_1 - x_2) = F \quad (2)$$

$$m_2\ddot{x}_2 + k_2(x_2 - x_1) = 0 \quad (3)$$

Suppose $x_1(0) = \dot{x}_1(0) = \dot{x}_2(0) = 0$. Is there a combination of $F \neq 0$ and $x_2(0)$ such that $x_1(t) = 0$ for $t \geq 0$?

Taking the Laplace transform of (3), we have

$$m_2 \left(s^2 X_2(s) - \underbrace{\dot{x}_2(0)}_{=0} - s x_2(0) \right) + k_2 X_2(s) = k_2 X_1(s).$$

The Laplace transform of (2) with zero initial conditions is

$$m_1 s^2 X_1(s) + k_d s X_1(s) + k_1 X_1(s) + k_2 X_1(s) = F(s) + k_2 X_2(s).$$

If we set $X_1(s) = 0$ in both equations, we have

$$m_2 (s^2 X_2(s) - s x_2(0)) + k_2 X_2(s) = 0$$

$$X_2(s) = \frac{s x_2(0)}{s^2 + \frac{k_2}{m_2}}$$

from (3) and

$$X_2(s) = \frac{-1}{k_2} F(s).$$

Combining, we have

$$F(s) = \frac{-s k_2 x_2(0)}{s^2 + \frac{k_2}{m_2}}.$$

Taking the inverse transform,

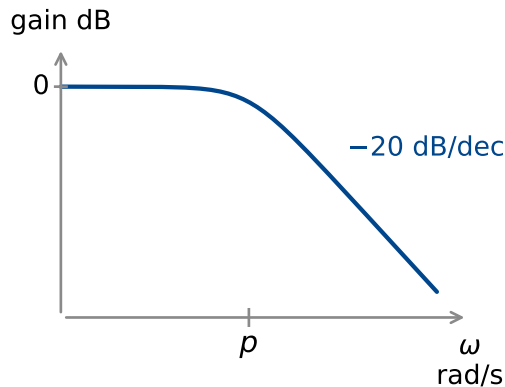
$$F(t) = -k_2 x_2(0) \cos \left(\sqrt{\frac{k_2}{m_2}} t \right).$$

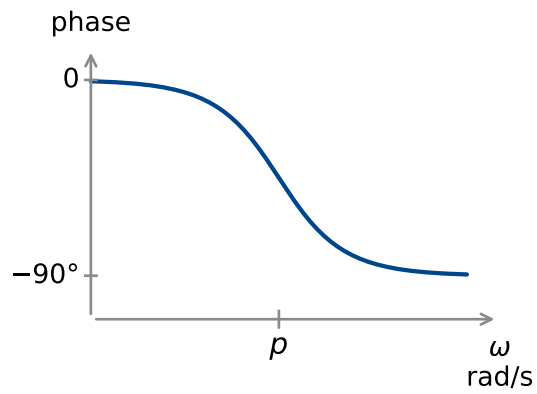
The extra mass completely rejects the sinusoidal input F , and the frequency depends only on m_2 , so it can be designed independently of m_1 . This idea is used in vibration damping, for example tuned mass dampers for earthquake protection in buildings and Stockbridge dampers on power lines.

Zeros in the frequency domain

In the frequency domain, zeros are the opposite of poles.

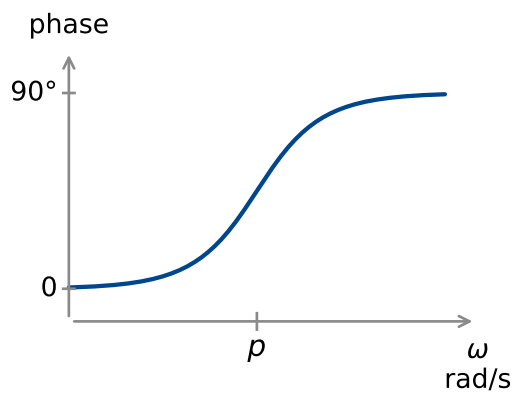
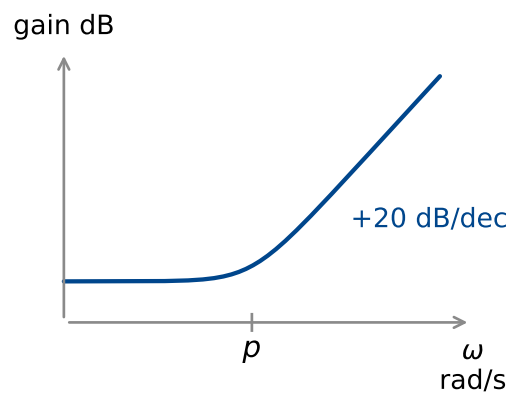
Poles give -20 dB/dec gain after the break frequency, and $-\frac{\pi}{2}$ total phase lag.



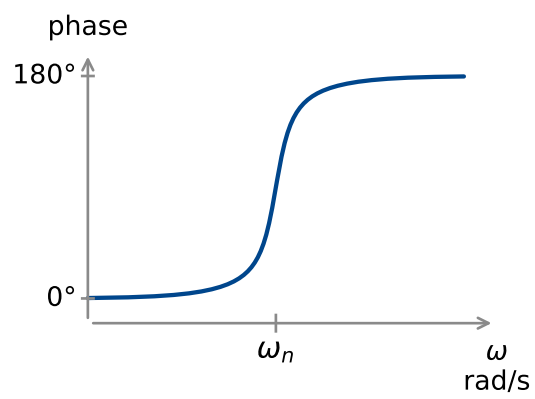
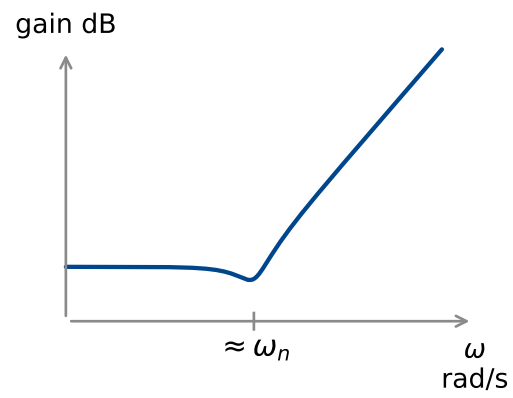


$$\frac{1}{s + p}$$

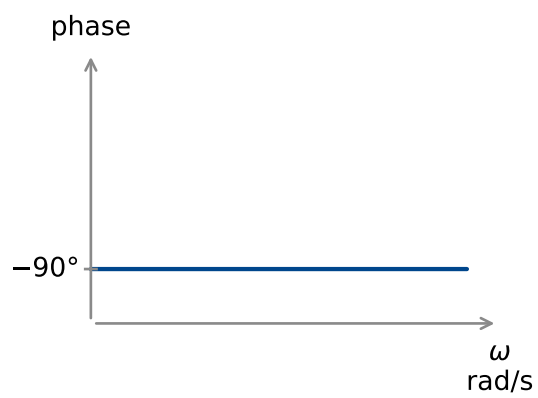
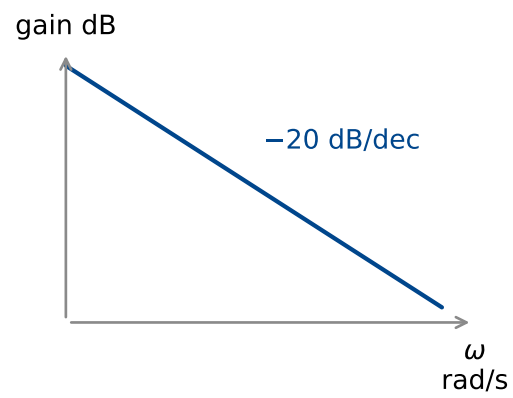
Zeros give +20 dB/dec after the break frequency, and $+\frac{\pi}{2}$ total phase lead.



Complex zeros give the opposite behaviour to complex poles.

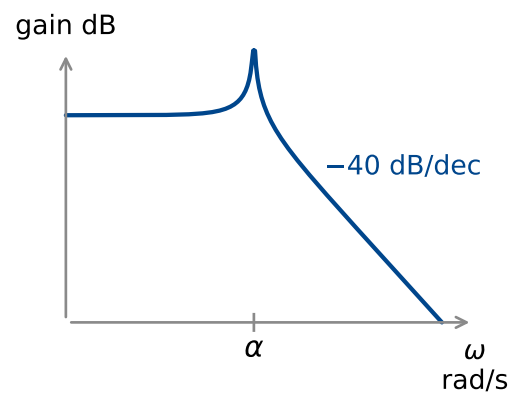
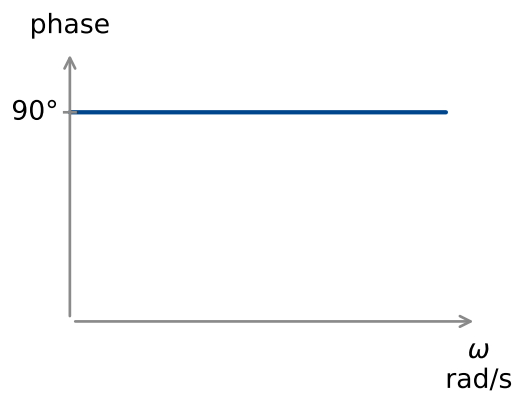
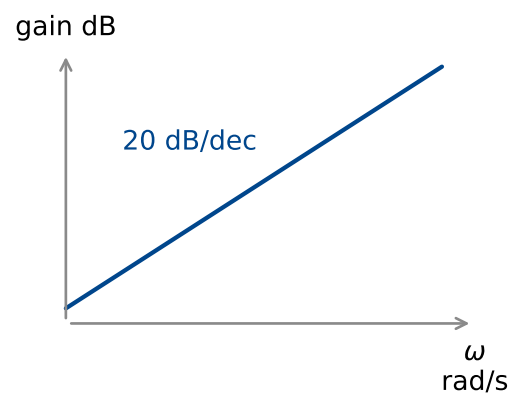


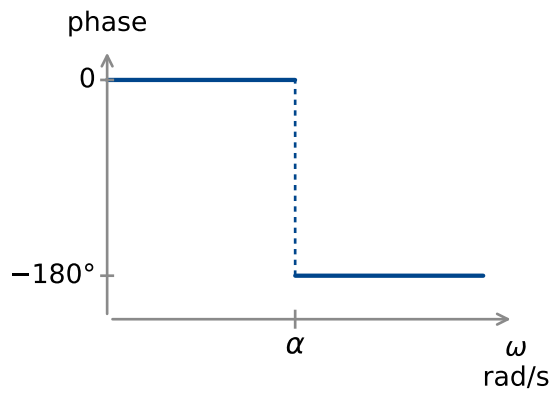
What about poles & zeros on the imaginary axis?



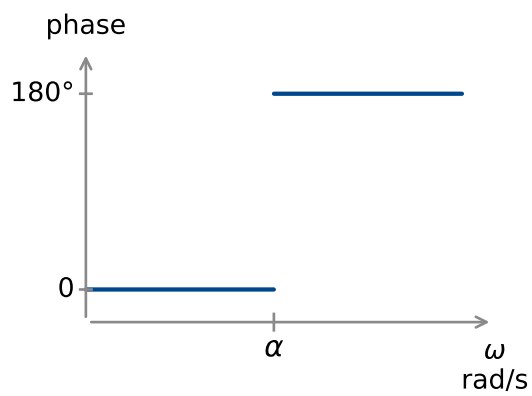
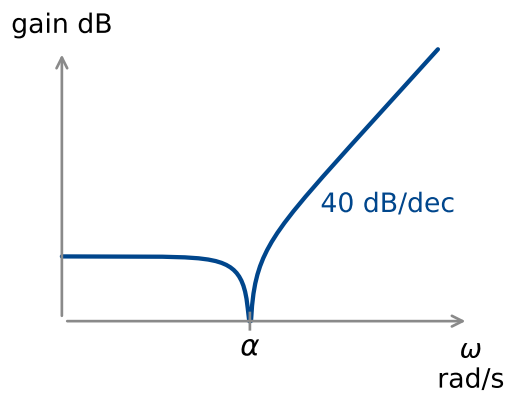
$$\frac{1}{s}$$

(integrator)





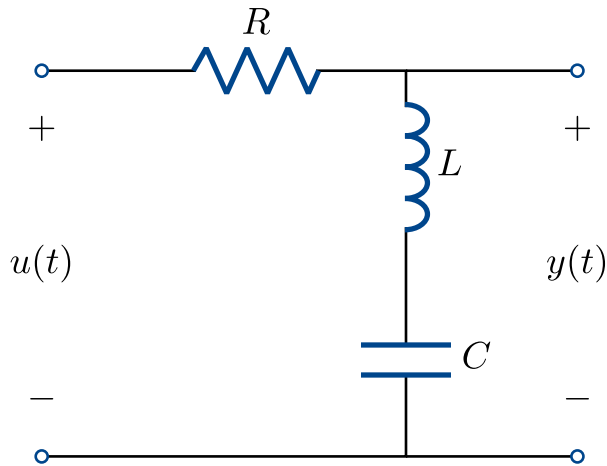
$$\frac{1}{(j\omega)^2 + \alpha^2}$$



$$(j\omega)^2 + \alpha^2.$$

We can use these to estimate the Bode diagrams of arbitrary transfer functions.

Example 8



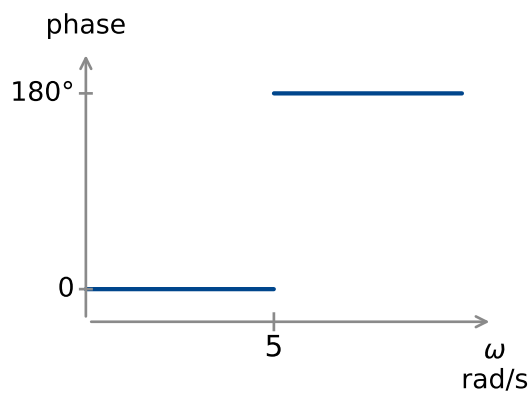
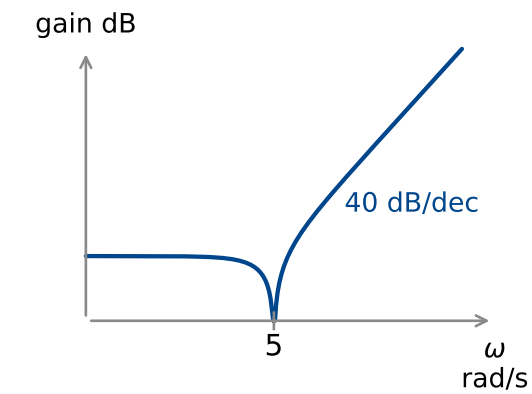
The frequency response of the series RLC with the output taken as the voltage across the capacitor and inductor is given by

$$G(j\omega) = \frac{(j\omega)^2 LC + 1}{(j\omega)^2 LC + j\omega RC + 1}$$

If we have $R = 6 \Omega$, $L = 1 \text{ H}$, $C = 40 \text{ mF}$, the transfer function is

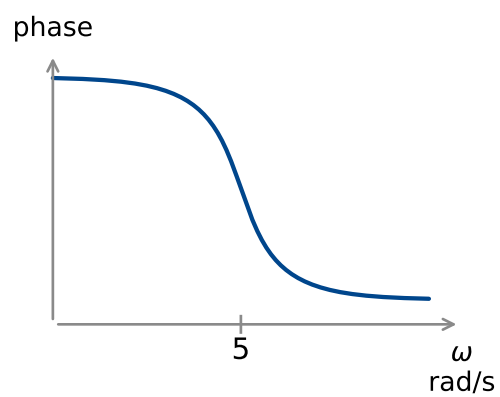
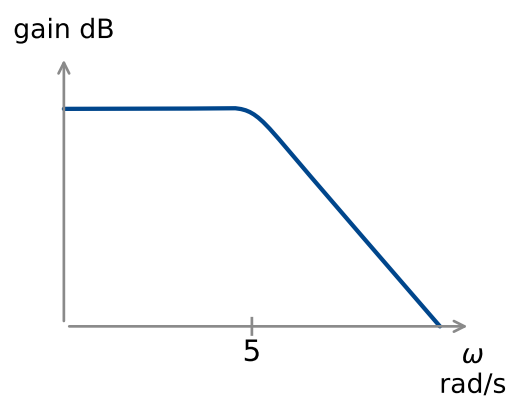
$$G(j\omega) = \frac{(j\omega)^2 + 25}{(j\omega)^2 + 6j\omega + 25}$$

Numerator:



$$(j\omega)^2 + 5^2.$$

Denominator:



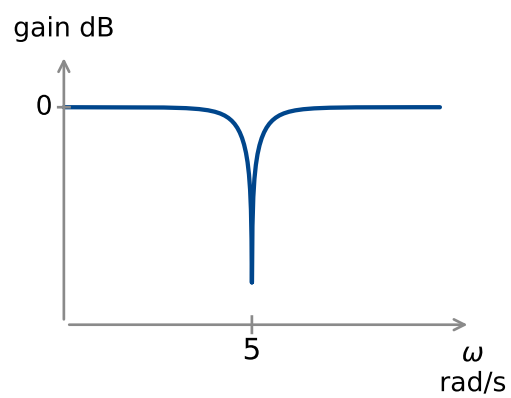
$$(j\omega)^2 + 6j\omega + 25$$

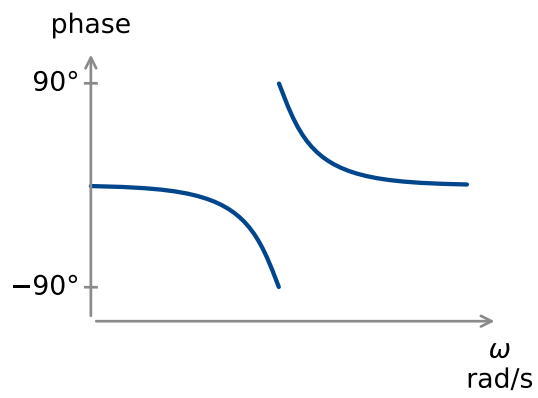
$$\omega_n = 5 \text{ rad/s,}$$

$$2\zeta\omega_n = 6$$

$$\zeta = \frac{6}{10} = 0.6.$$

Add together:





This is a notch filter – it excludes a narrow range of frequencies around 5 rad/s.
More on this next week when we look at filter design!